EECS 3311 – Software Design (Winter 2026)

Deliverable I – YorkU Lab Equipment Reservation Platform

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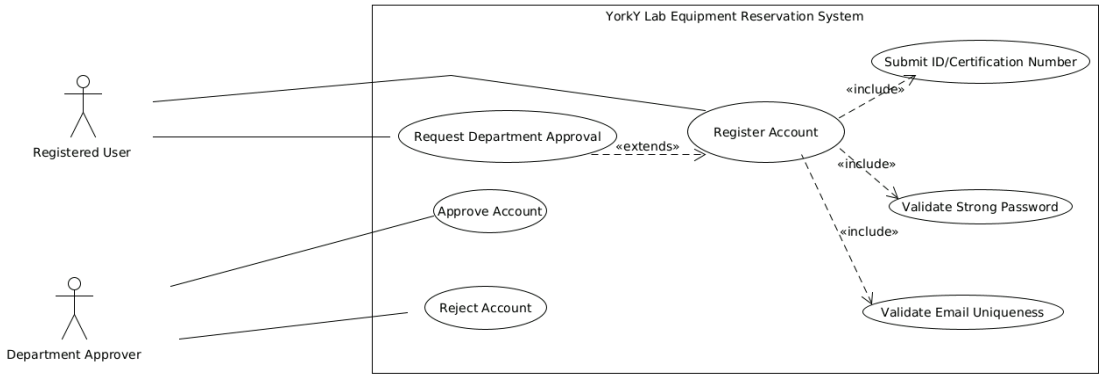
Github URL: https://github.com/Nahum-Durrani/EECS3311-Deliverable1-LabReservation

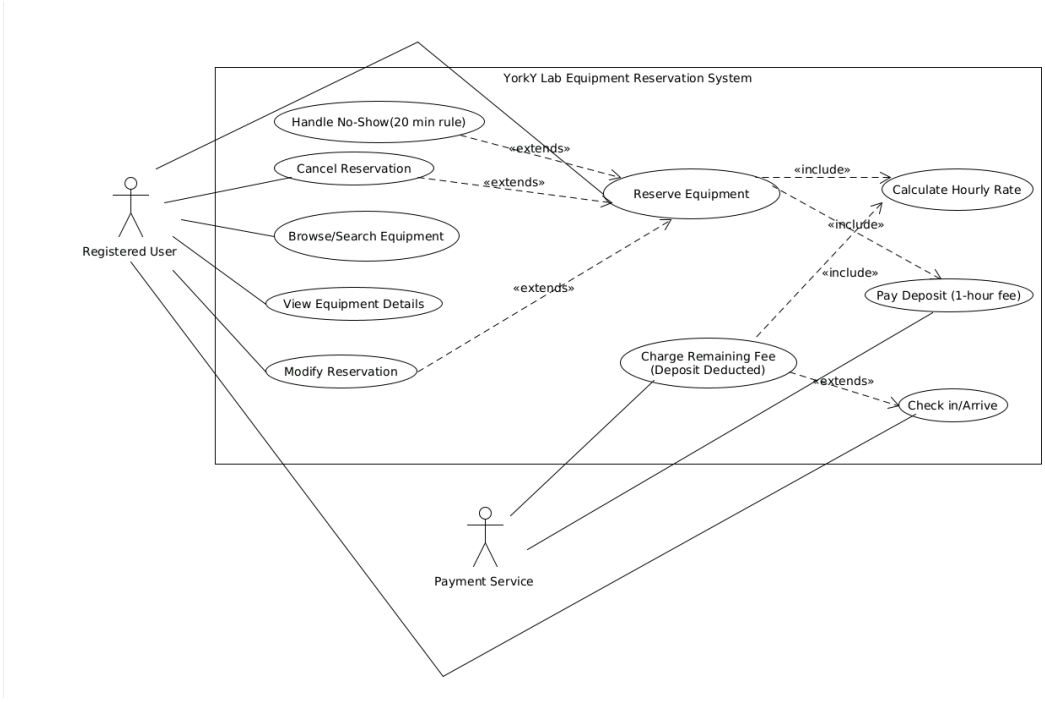
Part I – Requirements Modeling

Use Case Diagrams

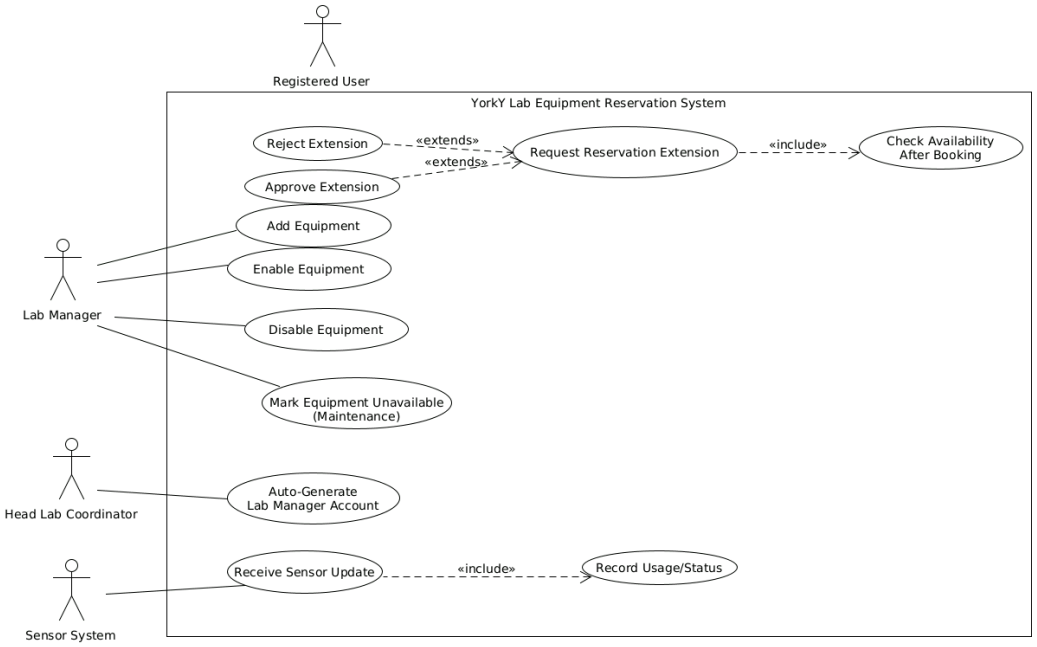
Use Case Diagram – Scenario 1 (Registration & Approval)



Use Case Diagram – Scenario 2 (Reservation + Deposit + No-Show)

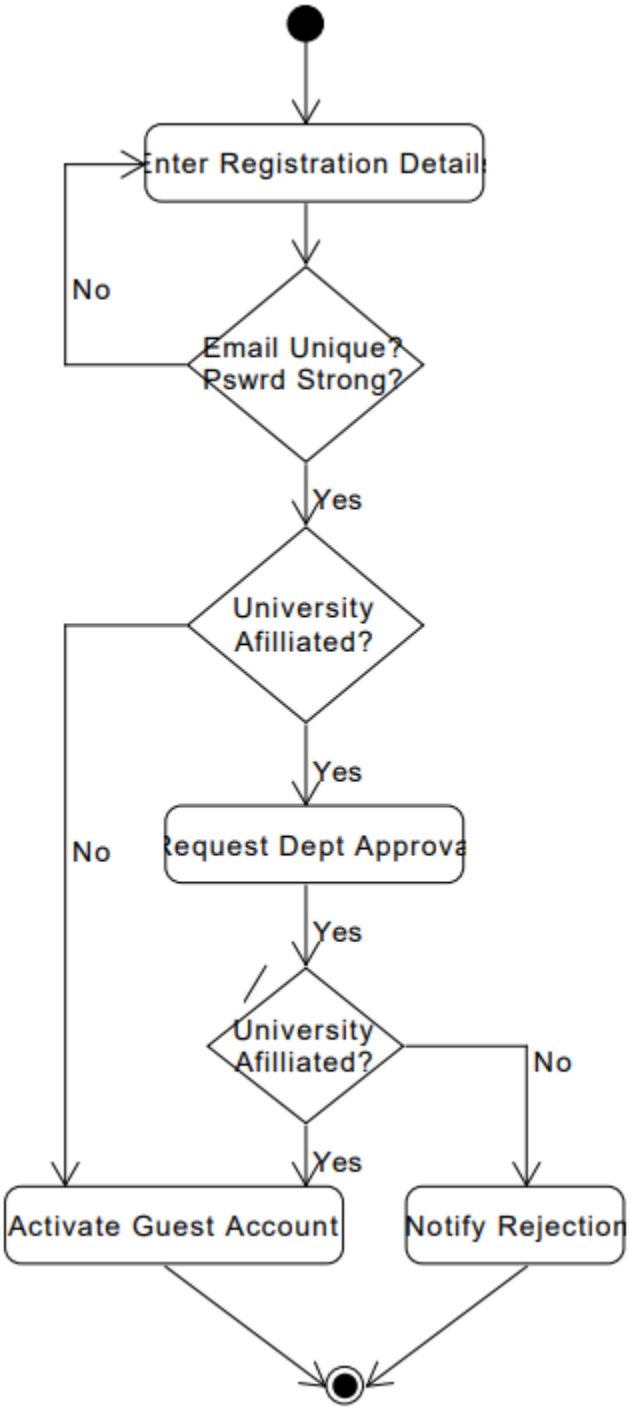


Use Case Diagram – Scenario 3 (Management + Sensors + Extension)

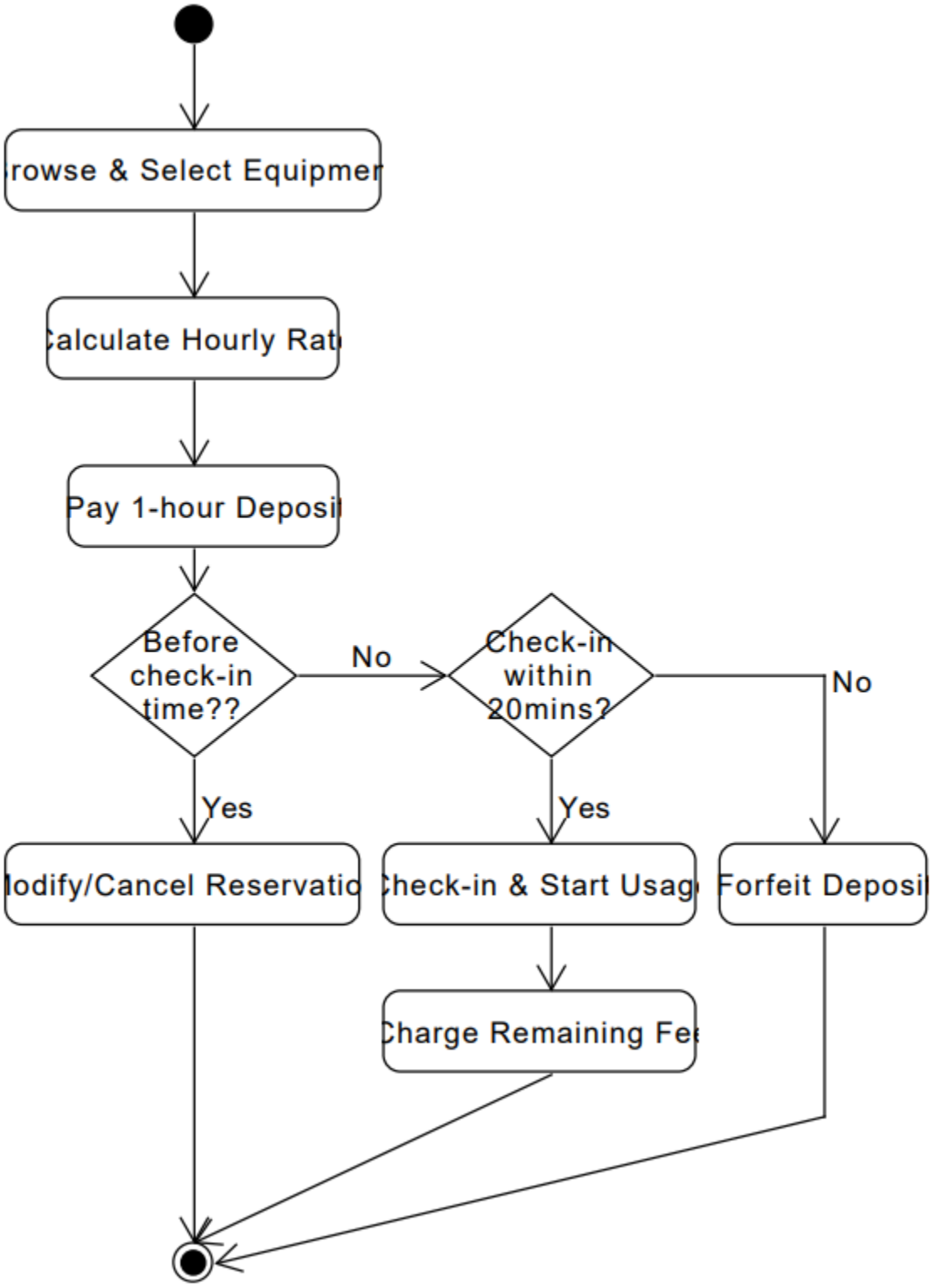


Activity Diagrams

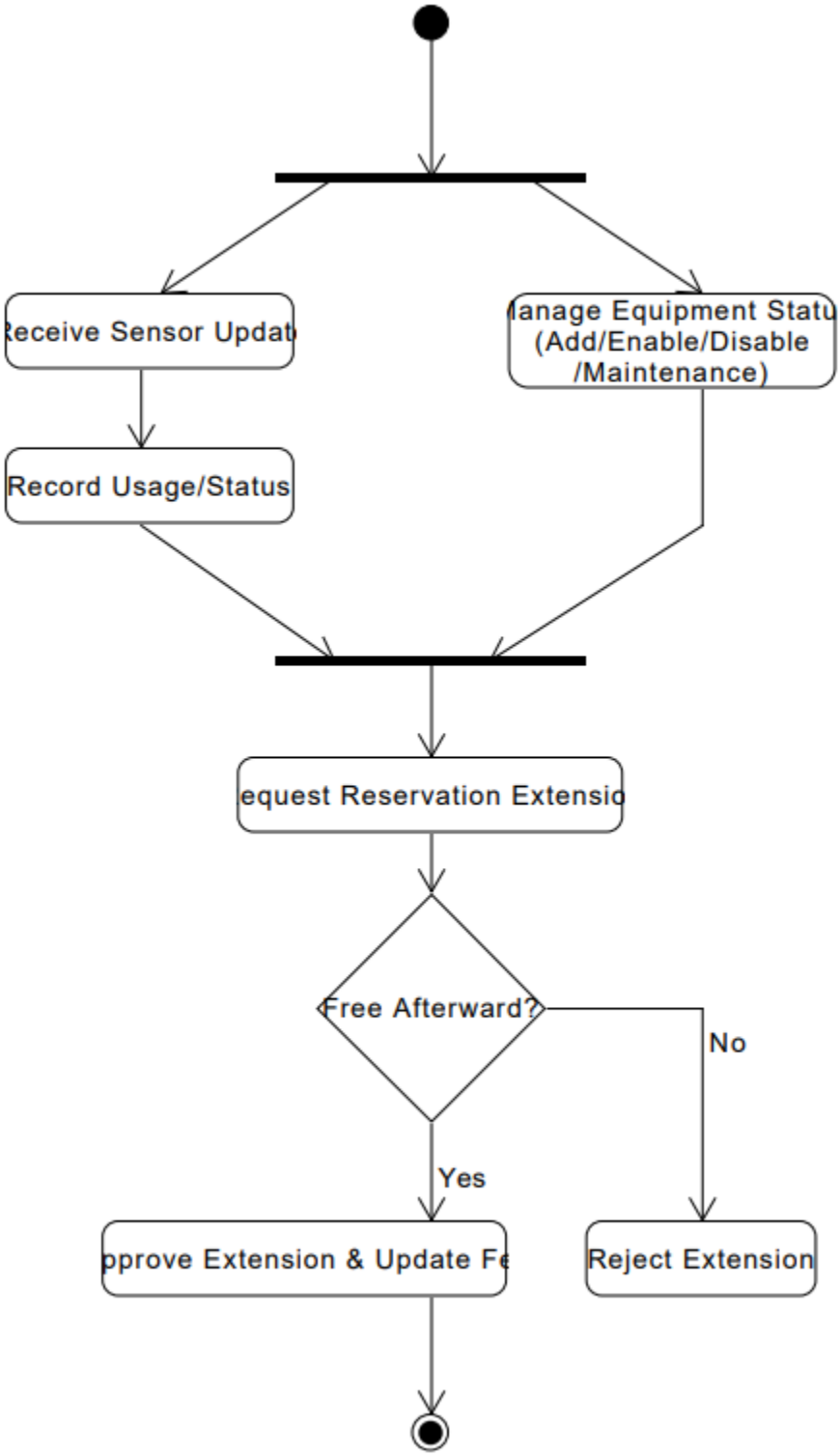
Activity Diagram – Scenario 1



Activity Diagram – Scenario 2

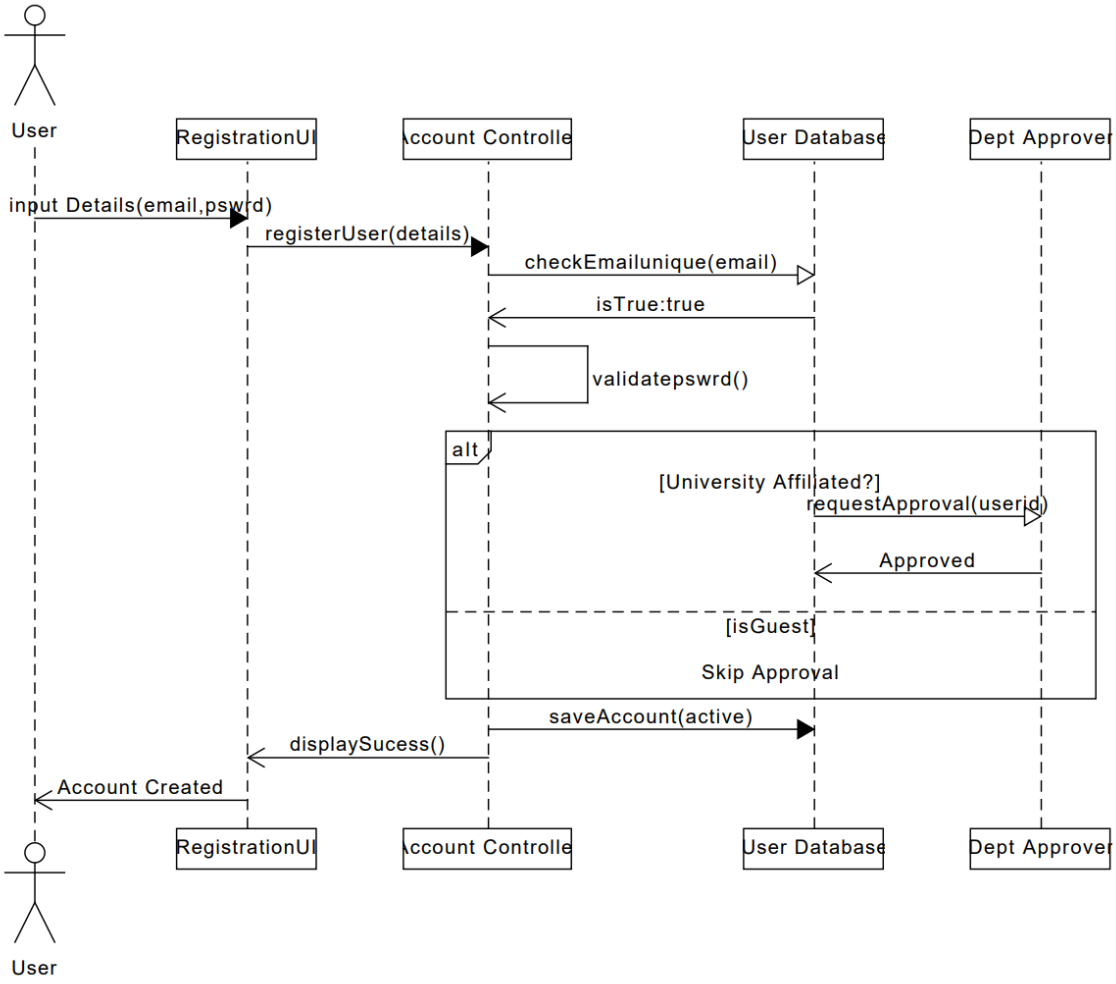


Activity Diagram – Scenario 3

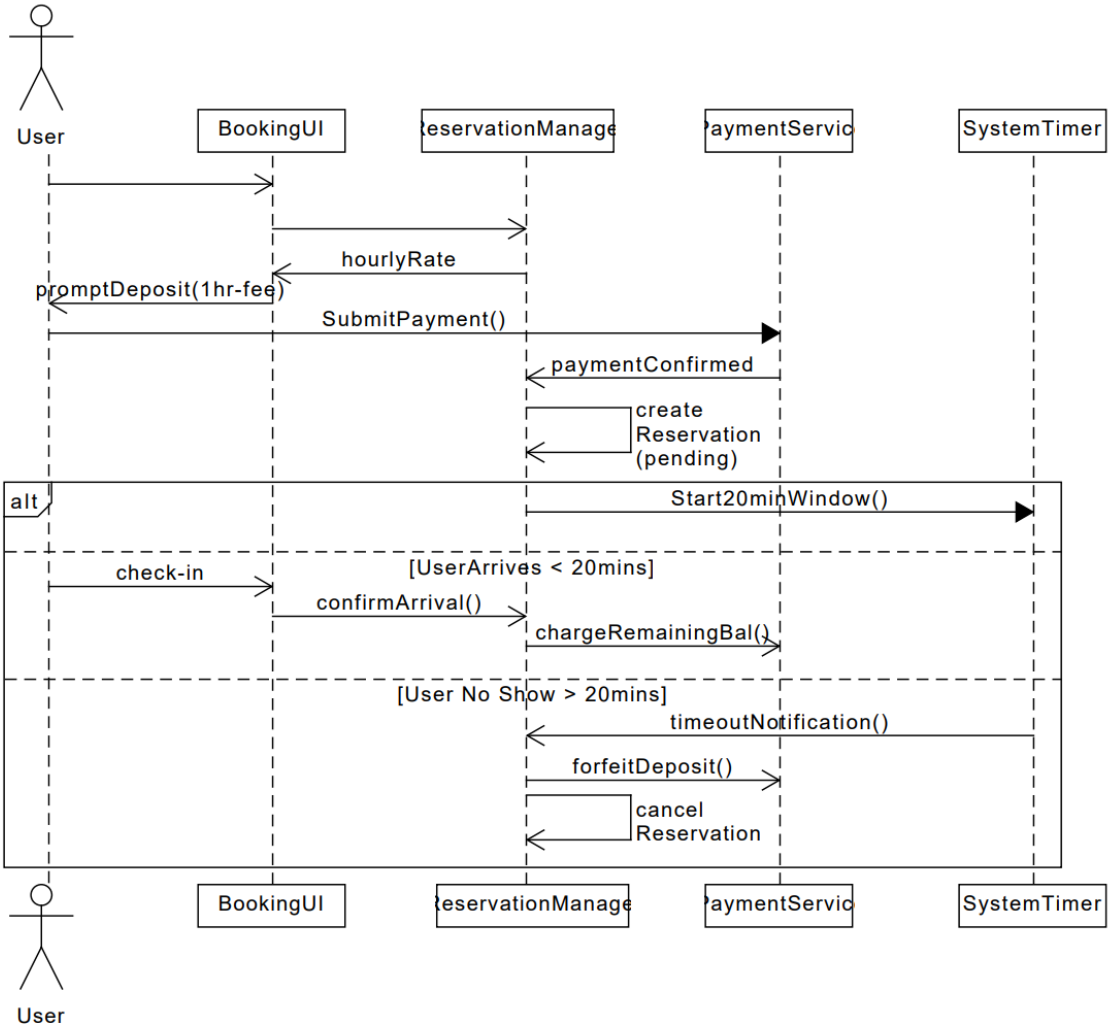


Sequence Diagrams

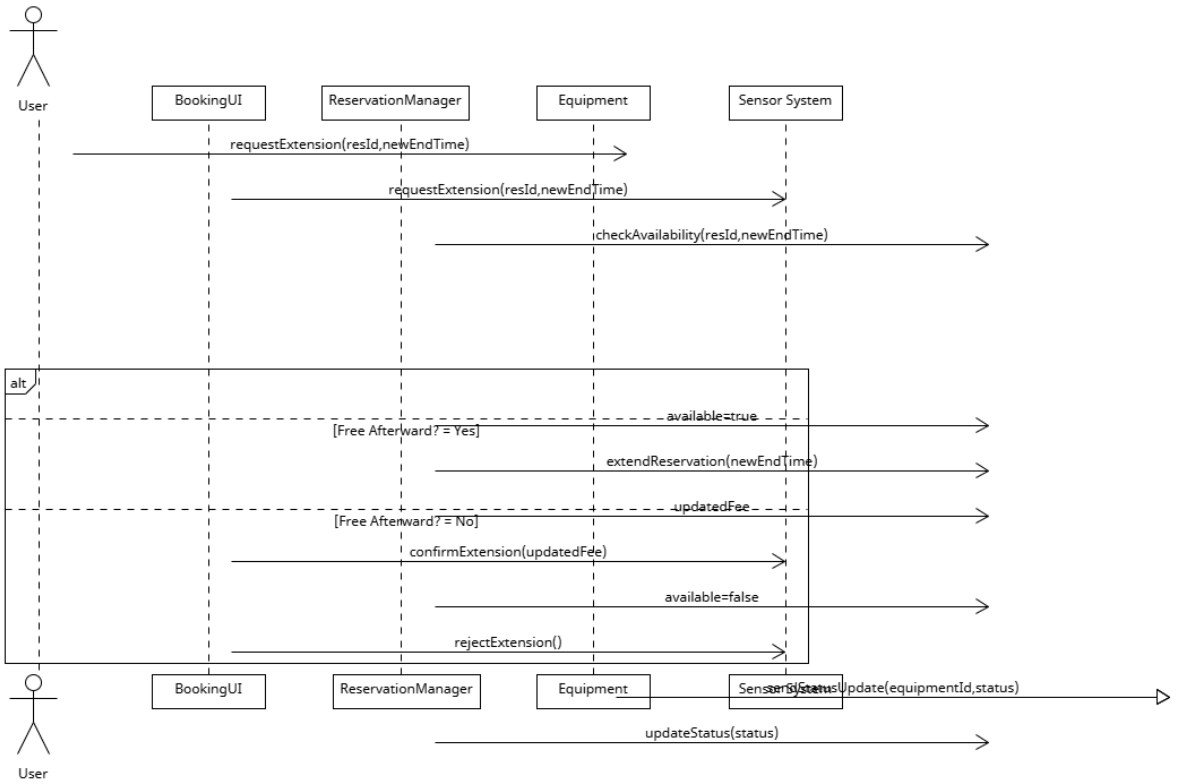
Sequence Diagram – Scenario 1



Sequence Diagram – Scenario 2

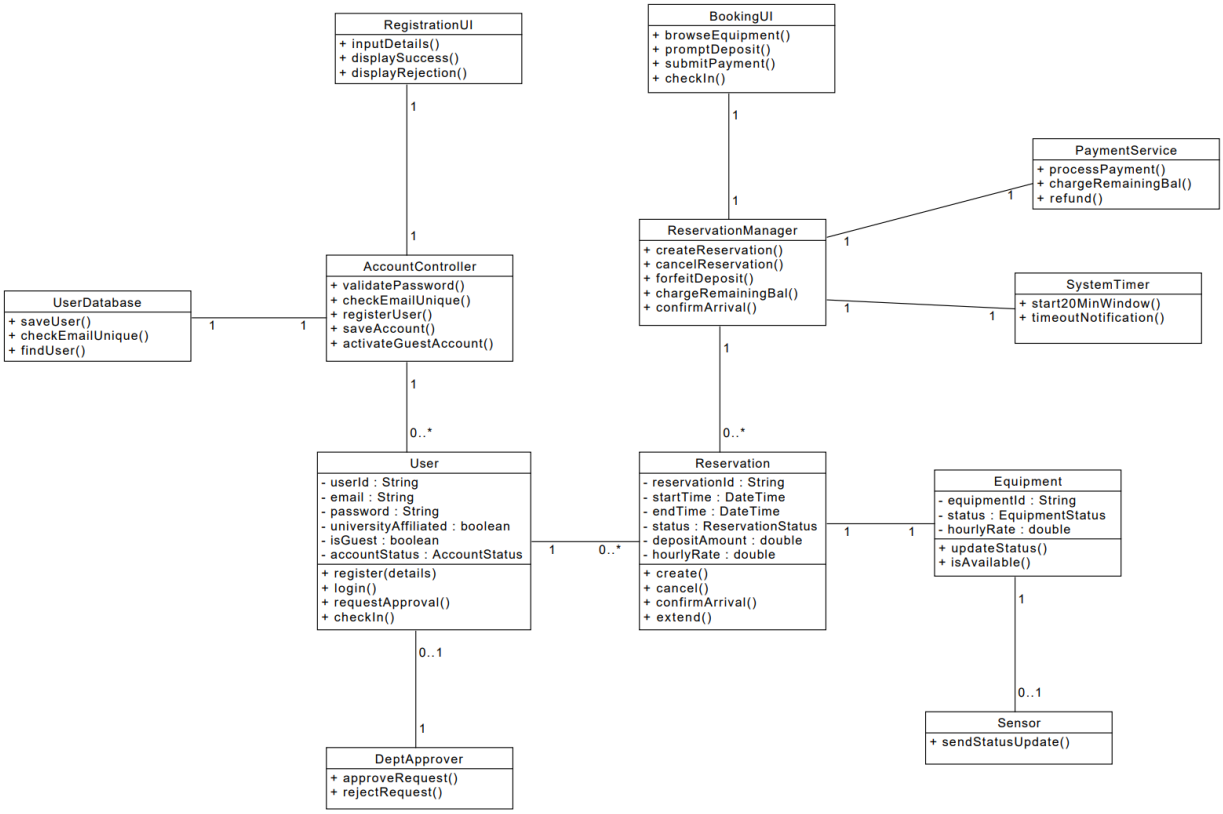


Sequence Diagram – Scenario 3



Part II – Design

Class Diagram



Part III – AI Assistant Adaptation Report

1. Role of the AI Assistant

Throughout Parts I and II of this project, AI tools were used to assist in drafting initial UML diagrams, identifying actors, and structuring system behavior. However, all AI-generated content was reviewed, verified, and refined to ensure alignment with project requirements (Req1–Req10) and UML standards.

1.1 Scenario 1 – Registration & Approval

AI Model Used

ChatGPT (GPT-5)

Prompt Used

Create a UML use case diagram for a university lab reservation system that includes strong password validation, unique email validation, ID submission, and conditional departmental approval.

AI Output Generated

The AI generated actors such as Registered User and Department Approver. It included use cases such as Register Account, Validate Strong Password, Validate Email Uniqueness, Submit ID/Certification Number, and Request Department Approval.

Understanding and Adaptation

The AI correctly identified validation requirements and ID submission. However, it initially modeled departmental approval as mandatory. Based on the requirements, approval is conditional and only applies to university-affiliated accounts. We corrected this by using an <<extend>> relationship instead of <<include>>. We also ensured the Approver actor had both Approve and Reject actions.

1.2 Scenario 2 – Reservation + Deposit + No-Show

AI Model Used

ChatGPT (GPT-5)

Prompt Used

Model a reservation system including hourly rate calculation, 1-hour deposit payment, 20-minute no-show forfeiture rule, and modify/cancel only before reservation start.

AI Output Generated

The AI generated use cases such as Reserve Equipment, Pay Deposit, Modify Reservation, and Cancel Reservation.

Understanding and Adaptation

The AI simplified the payment process and did not clearly represent the 20-minute no-show rule. We refined the activity diagram to include a decision node: If the user arrives within 20

minutes, the deposit is deducted; otherwise, it is forfeited. We also explicitly added the constraint that modification and cancellation are only allowed before the start time.

1.3 Scenario 3 – Management + Sensors + Extension

AI Model Used
ChatGPT (GPT-5)

Prompt Used
Generate UML use cases where only a Head Lab Coordinator creates manager accounts, managers maintain equipment, sensors send updates, and reservations can be extended only if available.

AI Output Generated
The AI identified actors such as Head Lab Coordinator, Lab Manager, and Sensor System. It included use cases for adding equipment, enabling/disabling equipment, marking maintenance, receiving sensor updates, and extending reservations.

Understanding and Adaptation
The AI initially did not strictly enforce that only the Head Lab Coordinator generates manager accounts. We corrected this by restricting that use case to that actor. We also ensured that extension only occurs if no scheduling conflict exists.

1.4 Class Diagram Design (Part II)

AI Model Used
ChatGPT (GPT-5)

Prompt Used
Design a class diagram for a lab equipment reservation system including users, reservations, equipment, payments, and sensors.

AI Output Generated
The AI suggested classes such as User, Reservation, Equipment, Payment, LabManager, and Sensor.

Understanding and Adaptation
The AI provided a good starting structure but lacked precise multiplicity. We refined associations to reflect that one User can have multiple Reservations, and each Reservation links to exactly one Equipment. Equipment exists independently from Reservation, so we modeled it using association rather than composition.

2. Arguments and Resolution

Discrepancies were identified between AI suggestions and system requirements:

• Department approval was incorrectly modeled as mandatory; corrected using <<extend>>.

• No-show forfeiture rule was not explicitly modeled; added conditional logic.

• Manager account creation restriction was not enforced; corrected to only Head Lab Coordinator.

• UML relationships were reviewed and corrected where necessary.

3. Verifying AI Responses

All AI outputs were verified against Req1–Req10 and UML standards from lecture materials. We ensured correct use of <<include>> and <<extend>>, validated multiplicity, and confirmed all constraints (20-minute rule, modify-before-start, role restrictions) were explicitly represented.

For example, the no-show requirement was cross-checked with the specification and explicitly implemented in the activity diagram.

4. Application of AI-Suggested Content

Prompt Used: Identify key classes for a lab reservation system including roles and payment logic.

AI Response (excerpt):

The system should include classes such as User, Reservation, Equipment, Payment, LabManager, and Sensor.

Where It Appears in Final Submission:

These classes appear directly in our final class diagram in Part II.

Rationale for Acceptance:

The class structure aligned with object-oriented design principles and required minimal modification.

Conclusion

AI accelerated the drafting process but required careful verification and refinement. The final submission reflects critical human evaluation of AI-generated content.